# LETTERS TO THE EDITORS

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## The Geomagnetic Field in Upper Triassic Times in the United States

It is commonly assumed[1] that magnetically stable rocks have a remanent magnetization in the direction of the geomagnetic field at the time of formation, and that the mean direction determined from samples collected through a considerable thickness of one rock formation is that of a dipole field orientated along the axis of rotation of the Earth at that time.

Because of the possibility of special effects, such as magnetostriction, the view has been taken[2] that these assumptions are likely to be in error and that in consequence the inferences which have recently been drawn from palæomagnetic studies[1] concerning the occurrence of polar wandering and continental drift in the geological past are based on no substantial grounds. A test relevant to this discussion is to determine whether rocks of the same age within one continent give pole positions in agreement, although it must be remembered that the geological age of those rocks which are most strongly magnetized, lavas and red sandstones, cannot usually be established with precision because diagnostic fossils are often scarce or lacking.

Measurements on rocks of Upper Triassic age collected in the United States have recently been made in Newcastle, Cambridge and Canberra, as shown in Table 1. In all cases the number of samples was small but they were well spread stratigraphically. The Springdale Sandstone[3] was collected from Zion National Park, Utah, and those from the eastern United States from the lower Connecticut Valley, from the upper Connecticut Valley and from New Jersey. The pole positions calculated on the assumption of an average dipole field are given in Table 1 and are plotted in Fig. 1. The ovals of confidence of the pole positions overlap.

The pole based on measurements of the Upper Triassic Keuper Marls of England[4] and its oval of confidence is also given for comparison—it is believed that this discrepancy results from post-Triassic continental drift.

From these results three points of interest arise.

(1) They show that results from small numbers of samples can be quite reliable and provide a practical demonstration of the validity of the sampling methods advocated by Irving and Watson[5] and by Runcorn[6].

(2) Results from both igneous and sedimentary rocks laid down at approximately the same geological time but at points thousands of miles apart within the same continent are in agreement, and indicate the high degree of certainty with which these 'fossil' directions may be identified with the Earth's magnetic field during the Upper Triassic so that in these red sandstones and lavas it seems that the special effects mentioned above are of negligible importance.

(3) It seems reasonably certain that during this time the geomagnetic field over the region of the United States approximated on average to that of a dipole as it is known to have done since the Miocene[7,8].

Table 1

| Formation | No. of determinations | Mean direction | | | Sampling area | | Pole position | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | $D$ | $I$ | $\alpha$ | Lat. | Long. | Lat. | Long. | $d\chi$ | $d\psi$ |
| 1. Springdale Sandstone | 18 disks from 8 samples | 338 | +16 | — | 37 N. | 113 W. | 55 N. | 107 E. | — | — |
| 2. Lavas near Holyoke, Mass. | 8 samples from 3 flows | 10 | +14 | 11° | 42 N. | 73 W. | 54 N. | 90 E. | 11 | 6 |
| 3. Lavas and sediments of Connecticut | 12 | 12 | +14 | 15° | 42 N. | 73 W. | 55 N. | 88 E. | 15 | 8 |
| 4. New Jersey (Brunswickian formation of Newark series) | 71 disks from 21 samples | 6 | +28 | 3° | 41 N. | 75 W. | 63 N. | 93 E. | 6 | 3 |

Mean directions of magnetization of Upper Triassic formations from the United States, specified by declination $D$ reckoned eastwards from geographical north and the inclination $I$ positive downwards. $\alpha$ is the angle of confidence at the 95 per cent probability-level. The semi-axes of the ovals of confidence of the poles, $d\psi$ and $d\chi$, in the direction of and perpendicular to the great circle joining the pole and the site are given. The significance of $\alpha$, $d\psi$ and $d\chi$ have been discussed previously (ref. 1).

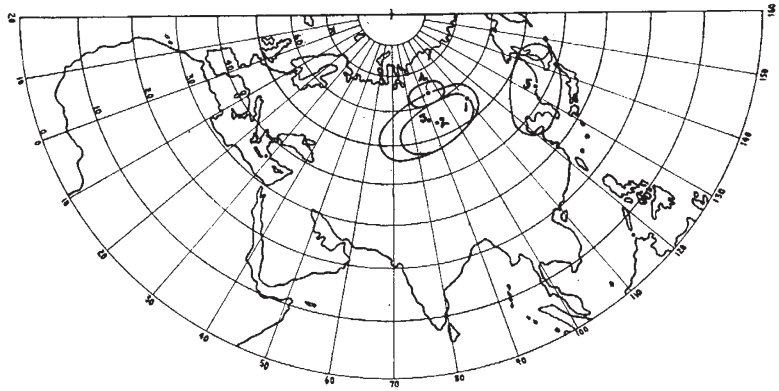


Fig. 1. Position of the geomagnetic pole in Upper Triassic times relative to the United States and Europe. The determinations from rock formation in the United States are numbered as in Table 1. The equivalent pole determined from England is numbered 5

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The axis of this dipole intersected the Earth's surface in the regions of what are now central Siberia and southern South America.

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## Charged Particles in a Non-uniform Radio-frequency Field

An analysis of the motion of a charged particle in a non-uniform radio-frequency field has been made and has shown that under certain conditions particles of either sign will experience an acceleration towards the position of least electric field strength.

Considering the Lorentz forces upon free electrons, a second-order force arises due to spatial variation of the electric field-strength, which is of the same order as the conventional forces due to motion in the magnetic field and charge density fluctuations. In a continuous but not necessarily homogeneous electron distribution, such as we suppose to exist in a collision-free and neutral plasma, a small-signal theory shows that the time-averaged force per unit volume upon the electrons is, $f = -\frac{1}{2}(\rho e/\omega^2 m)\nabla(\overline{E \cdot E})$, where $\rho$ is the average electron charge density, $\omega = 2\pi \times$ frequency, $\overline{E \cdot E}$ is time-averaged square of electric field, in rationalized M.K.S. units.

The total force upon an isolated body of electrons is the same as that given by conventional theory[1] from the field equations; the force is therefore consistent with the concept of radiation pressure. The internal stresses, or pressure in a fluid, differ, however, from the conventional theory, and appear to enhance the possibility of containing a heated plasma within a resonant cavity by radiation pressure[2].

It is hoped that it will shortly be possible to publish the details of the calculations which we have quoted.

The existence of such a force on electrons has been demonstrated in a 10-cm. slotted anode magnetron having no magnetic field and a cold cathode. Pulses of 10-cm. radio-frequency power were fed into this magnetron and set up electromagnetic fields which decreased towards the cathode. The multipactor effect produced a copious supply of electrons within the slots and limited the radio-frequency voltage there to about 5 kV.

Pulses of electrons arrived at the cathode with energies up to 250 volts, and current of up to 0·1 amp. per cm. length of cathode could be drawn with anode and cathode at the same potential. The anode and cathode diameters were 1·25 and 0·59 in. respectively. A small increase in cathode temperature was observed with a thermocouple, which proves that the current was due to negative particles bombarding the cathode.

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## Some Observations of the Infra-red Solar Spectrum from a High-flying Aircraft

The strong absorption bands of water vapour and carbon dioxide in the infra-red obscure much of the solar spectrum from observation on the ground. By working at high altitude, however, it is possible to use measurements of absorption in these bands for obtaining data about the concentration of water vapour and other atmospheric constituents in the upper atmosphere. To make such measurements, an infra-red grating spectrometer with a resolution $\sim 2$ cm.$^{-1}$ has been installed in a Canberra aircraft. A heliostat maintains a steady image of the sun on the input slit of the spectrometer. The spectrometer and heliostat are installed in the unpressurized part of the aircraft necessitating an elaborate system of remote control for its operation.

Successful ascents have been made in which the spectral region 1·8–2·1μ has been observed using a

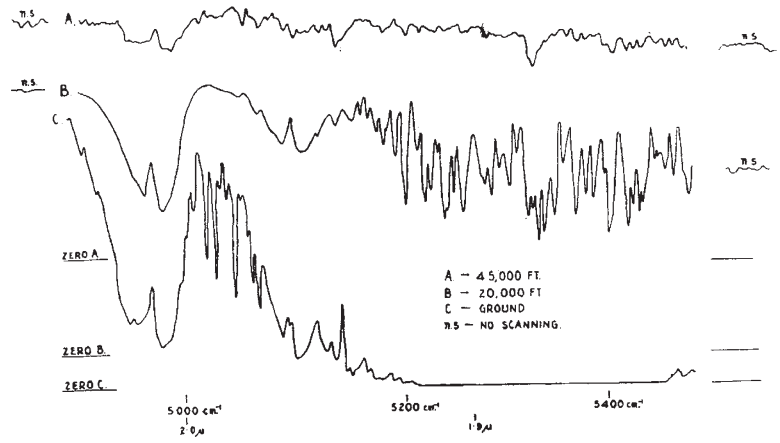



Fig. 1. Solar spectra

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